This is the condition for the point P to have a dark fringe. The distance $y_n$ is the distance of the $n^{th}$ dark fringe from the point $O$. The formation of bright and dark fringes is shown in Figure 7.14.

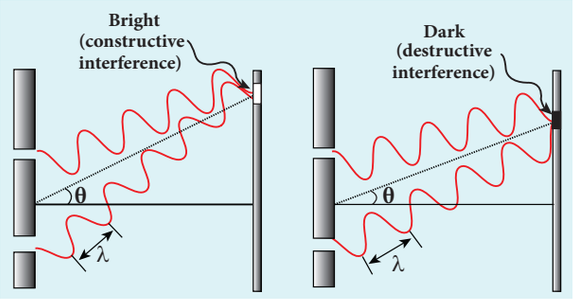



**Figure 7.14** Formation of bright and dark fringes

This shows that on the screen, alternate bright and dark fringes are seen on either side of the central bright fringe. The central bright is referred as $0^{th}$ bright followed by $1^{st}$ dark and $1^{st}$ bright and then $2^{nd}$ dark and $2^{nd}$ bright and so on, on either side of $O$ successively as shown in Figure 7.15.

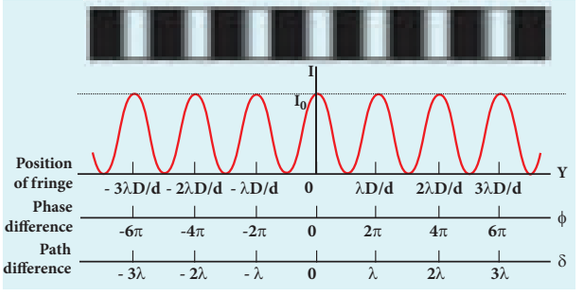



**Figure 7.15** Interference fringe pattern

*Equation for bandwidth*

**The *bandwidth* $\beta$ is defined as the distance between any two consecutive bright (or) dark fringes.**

The distance between $(n+1)^{th}$ and $n^{th}$ consecutive bright fringes from $O$ is given by,

$$\beta = y_{(n+1)} - y_n = \left((n+1)\frac{\lambda D}{d}\right) - \left(n\frac{\lambda D}{d}\right)$$

$$\beta \text{ for bright,} \quad \beta = \frac{\lambda D}{d} \qquad (7.31)$$

Similarly, the distance between $(n+1)^{th}$ and $n^{th}$ consecutive dark fringes from $O$ is given by,

$$\beta = y_{(n+1)} - y_n = \left(\frac{(2(n+1)-1)}{2}\frac{\lambda D}{d}\right) - \left(\frac{(2n-1)}{2}\frac{\lambda D}{d}\right)$$

$$\beta \text{ for dark,} \quad \beta = \frac{\lambda D}{d} \qquad (7.32)$$

From Equations (7.31) and (7.32) we understand that the bright and dark fringes are of same width equally spaced on either side of the central bright fringe.

*Conditions for obtaining clear and broad interference fringes:*

(i) The distance $D$ between the screen and double slit should be as large as possible.
(ii) The wavelength $\lambda$ of light used must be as long as possible.
(iii) The distance $d$ between the two slits must be as small as possible.

## EXAMPLE 7.6

In Young's double slit experiment, the two slits are 0.15 mm apart. The light source has a wavelength of 450 nm. The screen is 2 m away from the slits.

(a) Find the distance of the second bright fringe and also third dark fringe from the central maximum.

(b) Find the fringe width.

(c) How will the fringe pattern change if the screen is moved away from the slits?

(d) What will happen to the fringe width if the whole setup is immersed in water of refractive index 4/3.



UNIT-7(XII-Physics_Vol-2)_new.indd 65 21-07-2020 16:35:53